DR. THEJUS MAHAJAN

Research Software · Scientific Modelling · HPC / GPU

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PROFILE

Physicist (PhD) for research software and scientific modelling roles: simulation and data code that others can check. Available immediately; open to relocation anywhere in Germany.

- **Simulation code:** individual-based model in Fortran, particle loops parallelised with OpenMP; ported to Python/JAX for GPUs with output binary-compatible with the Fortran format.
- **Validated science:** the model reproduced the measured temperature optima of strains revived from sediment within the error bars.
- **Production data code:** registry pipeline in R (143,000+ records, ~300 clinics) cut from 1,834 to 1,349 lines, output byte-identical after every change.

2025–2026 career step: full-time retraining (Applied Biostatistics, grade 1.0), clinical-data internship, guest scientist at Hereon, HLRS HPC/AI course.

MODELLING RESEARCH

Postdoctoral Scientist, Marine Ecosystem Modelling

Universität Hamburg, Institute of Marine Ecosystem and Fishery Science

08/2021 - 01/2025 · Hamburg, Germany

- Built a 0-D Lagrangian individual-based model (IBM-CLC) of *Nodularia spumigena*: each cell carries its own temperature optimum (T_{opt}), growth follows a thermal performance curve, and dormant akinetes form and germinate with set probabilities. Manuscript in final preparation.
- Validation: reproduced the measured T_{opt} of strains revived from sediment (Medwed et al., 2024) within the error bars, working with the experimental colleagues.
- Traced a shift of about 1 °C in predicted T_{opt} to the random-number stream changing which akinetes germinated; the fix was standing genetic variation in germinating cells and a large akinete pool.
- Found that ERA5 sea-surface temperature, which follows the thin sun-warmed surface layer, under-represented the cold winter of 1987 in the water below; a Gaussian cold pulse from reported Baltic Sea temperatures brought the model to the experimental values.
- Parallelised the particle loops of the Fortran model with OpenMP; ported the simulation engine to Python/JAX for GPUs (Colab T4): branchless vectorised biology, output binary-compatible with the Fortran format.
- Also worked, less extensively, with the 1-D concentration-based formulation in GOTM-FABM; Linux cluster (SLURM, MPI); NetCDF/HDF5, xarray.
- Taught on the BSc course Marine Ecosystem Modelling (2022–23); MSc guest lecture “From 0D to 1D”. *Parental leave 01/2024 – 12/2024.*

Guest Scientist

Helmholtz-Zentrum Hereon, Ecosystem Modelling

05/2025 - 10/2025 · Geesthacht, Germany

Continued the Hamburg project after the contract ended, in parallel with the continuing education: finalised the modelling framework and prepared the simulation data for the IBM-CLC manuscript.

CLINICAL DATA AND STATISTICS

Biostatistics and Clinical Data Management (internship)

HealthTwiSt GmbH, under Dr. Andreas Busjahn

02/2026 - 04/2026 · Berlin, Germany

Service company for the statistical analysis of clinical, genetic and scientific studies.

- Refactored the production R pipeline of the DeGIR[1] quality registry for

[1] DeGIR: Deutsche Gesellschaft für Interventionelle Radiologie und minimal-invasive Therapie.

SKILLS

Modelling
Individual-based (agent-based) models, stochastic simulation, thermal performance curves, validation against experimental data, GOTM-FABM, ERA5 forcing

Statistics
Linear, logistic and mixed models, multivariate methods, MCMC, random forests, boosting (coursework, grade 1.0)

Programming
Python (NumPy, Pandas, JAX, PyTorch, PySpark, scikit-learn), R (tidyverse, tidymodels, Shiny), Fortran, SQL, Bash

Computing
OpenMP, GPUs (Colab T4, V100), PyTorch multi-GPU (DDP, 4 GPUs), Spark MLlib, PBS batch jobs (HLRS training cluster), Linux cluster (SLURM, MPI)

ML & AI
PyTorch, Spark MLlib, LLM agents with tool calls (MCP, LangGraph), answers checked against SQL ground truth

Code quality
Git, automated tests, regression checks against reference output, R packages (roxygen2, devtools), Quarto

HPC AND AI TRAINING

Deployable Data Analysis & AI Pipelines with HPC

Supercomputing-Akademie, HLRS Stuttgart, 09–10/2026. Coursework and final examination completed 16.10.2026; certificate pending.

- PBS batch and interactive jobs on GPU (V100) and CPU (Skylake) nodes.
- Same network trained on both node types: 5.3 against 25.3 s per epoch (4.8×), with the reasons the gain is not the hardware ratio.
- Multi-GPU training with PyTorch DDP on 4 GPUs (course assignment).

Parallel Programming with MPI

HLRS Stuttgart. Registered; 02.11.–04.12.2026.

IBM: Deep Learning with PyTorch

Coursera, completed 08/2026.

Introduction to Supercomputing

Jülich Supercomputing Centre, 11/2025.

OPEN-SOURCE PROJECT

Neural-network interpretability

07/2026 – present

Python/PyTorch analysis of the Leela Chess Zero network (github.com/thejusmahajan/lc0-interpretability). 345 automated tests, key guards mutation-checked. Found two silent errors in my own analysis, one after publication; corrected both publicly.

LANGUAGES

interventional radiology (143,000+ records, ~300 clinics) from 1,834 to 1,349 lines; output verified byte-identical with `identical()` after every change.
- Checked patient-level records; defined clinical endpoints (success / partial success / failure) for 26 intervention types.
- Moved 241 data-correction rules and 16 success criteria out of the code into configuration files. Found and documented two existing errors; behaviour left unchanged.
- R package `degirtools` (9 functions, roxygen2, `devtools::check()` with 0 errors); moved ML teaching scripts from caret to tidymodels.

Continuing Education: Applied Bioinformatics & Biostatistics
CQ Beratung + Bildung GmbH, full-time
08/2025 - 02/2026 Berlin, Germany

Completed 04/2026 (with the internship). Module grades 1.0 – 2.1.
- Applied Biostatistics in R, grade 1.0: linear models with interactions, logistic regression, linear mixed models (random intercept and slope), PCA, discriminant and cluster analysis, MCMC (Metropolis), random forests and boosting.
- Programming in Python (1.2); Bioinformatics & Databases, SQL/MySQL (1.4); final project: a Nextflow (DSL2) pipeline.

English ●●●●●
Fluent (C1)

German ●●●○○
B1 (Goethe-Zertifikat), B2 exam booked for 10.12.2026

French ●●○○○
Basic

Malayalam ●●●●●
Native

Tamil, Hindi ●●●○○
Working proficiency

EARLIER CAREER

Doctoral Researcher
Université Paris-Saclay, ISMO
10/2015 - 09/2018 Orsay, France

- Collision experiments on the AGAT setup at the Orsay Tandem accelerator: data acquisition, calibration and quality control of multi-detector data.
- Classical Trajectory Monte Carlo (CTMC) calculations at Universidad Autónoma de Madrid and Université de Bordeaux; branching ratios contributed to the KIDA astrochemistry database.

Physics Tutor & Chess Coach
Freelance / Tutorwaves Solutions Inc.
10/2018 - 03/2021 Kerala, India

One-to-one teaching in physics and chess.

Relocation to Germany
04/2021 - 07/2021 Hamburg, Germany

Move from India to Germany; job search.

EDUCATION

Ph.D. in Lasers, Molecules and Atmospheric Radiation
Université Paris-Saclay
10/2015 - 09/2018 Orsay, France

Thesis applied to astrochemistry: *Excitation and fragmentation of C_nN^+ (n=1–3) in collisions with He atoms*.

M.Sc. in Physics
National Institute of Technology Calicut
07/2012 - 12/2014 Calicut, India

First Class with Distinction (CGPA 8.71/10). B.Sc. in Physics, University of Calicut, 2009–2012.

PUBLICATIONS

Four peer-reviewed journal articles (*Physical Review A*; *J. Phys. B: At. Mol. Opt. Phys.*; *Astronomy & Astrophysics*; *Molecular Astrophysics*) and two refereed conference papers (*J. Phys.: Conf. Ser.*), 2017–2020; first author on three. One modelling manuscript (IBM-CLC) in final preparation.

REFEREES

Available on request.